Bronchiectasis and Nasal Polyposis (BENP): A Comprehensive Disease Report

Summary

Bronchiectasis and Nasal Polyposis (BENP; OMIM:620984, MONDO:0975835, MedGen:1874999, UMLS:C5975469) is a distinct, recently-defined autosomal-recessive Mendelian airway disease caused by biallelic loss-of-function of *WFDC2* (WAP four-disulfide core domain 2; also known as HE4, human epididymis protein 4). WFDC2 is a small secreted WAP-domain protease inhibitor produced by airway secretory (club) cells and submucosal glands. When both copies of the gene are non-functional, the protein is not secreted into the airway surface liquid, removing a component of innate airway defense. The clinical consequence is chronic airway infection producing severe rhinosinusitis, pronounced nasal polyposis, and bronchiectasis — while ciliary structure/function and CFTR function remain normal. The disease was molecularly defined by Dougherty et al. in 2024 (PMID: 38626355), who identified biallelic pathogenic *WFDC2* variants in 11 individuals from 10 unrelated families across the United States, Europe, Asia, and Africa. A recurrent founder missense variant, p.Cys49Arg, disrupts glycosylation and blocks secretion of mature WFDC2.

This report distinguishes the molecularly-defined Mendelian entity **BENP (WFDC2 deficiency)** from the historically-described, clinically-overlapping **Woakes' syndrome** — a rare hereditary chronic rhinosinusitis with nasal polyposis (CRSwNP) variant defined by a clinical pentad (recurrent nasal polyposis, nasal broadening, frontal sinus aplasia, bronchiectasis, and dyscrinia). Woakes' syndrome, first delineated in 1979 (PMID: 553887), has never been assigned a molecular cause and has no OMIM/MONDO/Orphanet ID; it likely represents a clinically-recognized subset of the broader phenotypic spectrum that BENP now molecularly explains for a proportion of cases. Both entities share the core "unified airway" pathology linking upper-airway polyposis to lower-airway bronchiectasis via impaired mucociliary clearance and chronic infection.

Because WFDC2 (HE4) is undetectable in the serum of affected individuals, a **blood HE4 test combined with *WFDC2* genetic testing enables diagnosis** — a rare instance of a simple, accessible biomarker for a genetic airway disease. Management combines endoscopic sinus

surgery (universal in reported Woakes' cases), type-2-targeted biologics for the polyp component (dupilumab is the leading agent), and standard bronchiectasis care (airway clearance, anti-inflammatory DPP-1 inhibition, and infection control).

Disease Information

Overview

BENP is an **autosomal recessive airway disease characterized by chronic infection of the airways resulting in rhinosinusitis and pronounced nasal polyposis** (MedGen definition, after Dougherty et al., 2024). The disease spans the entire respiratory tract: upper airway (chronic rhinosinusitis, nasal polyposis, frequently with nasal broadening and frontal sinus abnormalities) and lower airway (bronchiectasis with chronic productive cough, recurrent infections, and progressive bronchial dilatation).

Key Identifiers

Resource	Identifier
OMIM	620984
MONDO	MONDO:0975835
MedGen	1874999
UMLS	C5975469
Causal gene (NCBI Gene)	*WFDC2*, Gene ID 10406
Cytogenetic locus	20q13.12
HGNC	HGNC:15466
Gene alias	HE4; the *WFDC2* alias list literally includes "BENP"
Related historical entity	Woakes' syndrome (no OMIM/MONDO/Orphanet ID; closest MedGen concept "Polypoid sinus degeneration", UMLS C0155822)

Synonyms and Alternative Names

- Bronchiectasis and nasal polyposis (BENP)
- Recessively inherited WFDC2 (HE4) deficiency
- (Clinically overlapping) Woakes' syndrome — synonyms: necrotic ethmoiditis with nasal polyposis; Woakes' ethmoiditis

Source of Information

Information is derived from **aggregated disease-level resources** (OMIM, MONDO, MedGen) and from **primary clinical genetics literature** (a multi-family international case series, Dougherty et al., 2024). Epidemiologic descriptors of the overlapping Woakes' phenotype come from a PRISMA systematic review of published individual cases (PMID: 41416018).

Etiology

Disease Causal Factors

BENP is a monogenic (Mendelian) disease. The primary and sufficient cause is **biallelic (homozygous or compound heterozygous) loss-of-function of *WFDC2***. As stated by Dougherty et al.: *"We identified biallelic pathogenic variants in WAP four-disulfide core domain 2 (WFDC2) in 11 individuals from 10 unrelated families originating from the United States, Europe, Asia, and Africa"* and *"WFDC2 dysfunction defines a novel molecular etiology of bronchiectasis characterized by the deficiency of a secreted component of the airways"* (PMID: 38626355).

A secondary infectious/mechanistic contribution is inherent: loss of the WFDC2 airway-defense protein permits **chronic bacterial colonization and infection**, which drives the inflammatory and structural airway damage.

Risk Factors

Genetic risk factors - Causal variants: biallelic pathogenic *WFDC2* variants. The recurrent founder missense variant **p.Cys49Arg** is the best-characterized; computer simulations and deglycosylation assays indicate it *"structurally"* disrupts glycosylation and blocks secretion of the mature protein (PMID: 38626355). - Susceptibility for the broader (non-Mendelian) nasal-polyp phenotype: CRSwNP has a strong heritable component. A Utah genealogical study (1,638 CRSwNP probands) found first-degree relatives had a **4.1-fold increased risk** ($p < 10^{-3}$) and second-degree relatives a 3.3-fold risk of the same diagnosis, with **no increased risk in spouses**: *"First-degree relatives (1stDRs) of CRSwNP patients demonstrated a 4.1-fold increased risk (p < 10(-3)) of carrying the same diagnosis, whereas second-degree relatives (2ndDRs) demonstrated a 3.3-fold increased risk"* and *"No increased risk was observed in spouses of CRSwNP patients"* (PMID: 25677865).

Environmental risk factors - For BENP specifically, environment is not a primary driver, but chronic airway infection (opportunistic bacterial colonization) perpetuates disease. In the reported Woakes' aggregate, disease was **male-predominant (65%)** with childhood-through-adult onset (PMID: 41416018). - Consanguinity increases the likelihood of recessive disease (relevant to a recessive disorder like BENP); in pediatric non-CF bronchiectasis cohorts consanguinity was positive in 59.4% (PMID: 29605210).

Protective Factors

No specific genetic or dietary protective factors are established for BENP. Heterozygous carriers of a single pathogenic *WFDC2* allele are unaffected (recessive inheritance), implying one functional allele is protective/sufficient.

Gene–Environment Interactions

The core interaction is **genotype (WFDC2 loss) × airway microbial exposure**: absent WFDC2-mediated innate defense, ordinary airway microbial exposure produces chronic infection and the self-sustaining "vicious vortex" of bronchiectasis. This is inferred from the protein's antibacterial function and the infection-dominated phenotype rather than directly demonstrated with GxE modeling.

Phenotypes

Phenotype	Type	HPO suggestion	Onset	Severity/ Course	Frequency
Nasal polyposis (recurrent, severe)	Clinical sign / physical manifestation	HP:0100582 (Nasal polyposis)	Childhood (classically early) to adult	Severe, recurrent/ relapsing	Defining feature; universal
Bronchiectasis	Clinical sign / imaging	HP:0002110 (Bronchiectasis)	Childhood–adult	Progressive, chronic	Defining feature
Chronic rhinosinusitis	Symptom/ sign	HP:0000246 (Sinusitis)	Childhood–adult	Chronic, relapsing	Very frequent
Chronic productive cough / sputum	Symptom	HP:0031245 (Productive cough)	With bronchiectasis onset	Chronic	Frequent
Recurrent respiratory infections	Symptom	HP:0002205 (Recurrent respiratory infections)	Childhood	Recurrent	Frequent
Nasal broadening	Physical manifestation	HP:0000414 (Broad nasal tip)	Childhood	Progressive with polyp mass	Woakes' feature
Frontal sinus aplasia	Physical/ imaging	HP:0002688 (Aplasia of the frontal sinus)	Congenital/ developmental	Stable	Woakes' feature
Dyscrinia (highly viscous mucus)	Laboratory/ physical	HP:0011950	Early	Chronic	Woakes' feature
Anosmia/ hyposmia	Symptom	HP:0000458 (Anosmia)	With polyp burden	Fluctuating	Frequent in CRSwNP

Clinical characteristics. The Woakes' pentad — *"recurrent nasal polyposis with broadening of the nose, frontal sinus aplasia, bronchiectasis, and dyscrinia (production of highly viscous mucus)"* (PMID: 553887) — captures the phenotype. A modern series reiterates *"severe recurrent nasal polyps in early childhood with broadening of the nose, nasal dyscrinia, frontal sinus aplasia and bronchiectasis"* (PMID: 27143164). Age of onset is variable — the systematic review of Woakes' cases found mean age 39.5 years (range 5–81), with both paediatric- and adult-onset cases (PMID: 41416018).

Quality-of-life impact. Substantial. The nasal-polyp component causes nasal obstruction, anosmia, and rhinorrhea (measured by SNOT-22); the bronchiectasis component causes chronic cough, sputum, dyspnea, and recurrent exacerbations. Endoscopic sinus surgery improves symptoms and QoL, but *"recurrences are common and multiple surgeries are often required"* (PMID: 41261359).

Genetic / Molecular Information

Causal Gene

WFDC2 (WAP four-disulfide core domain 2 / HE4 / human epididymis protein 4); **NCBI Gene 10406**; **HGNC:15466**; UniProt Q14508; chromosome **20q13.12**. OMIM phenotype 620984.

Pathogenic Variants

- **Variant type/class:** predominantly **missense** (recurrent founder **p.Cys49Arg**), consistent with disruption of a disulfide-bonded WAP-domain cysteine. Other biallelic pathogenic/likely-pathogenic variants were identified across the 10 families.
- **Classification (ACMG/AMP):** pathogenic/likely-pathogenic biallelic variants in affected individuals.
- **Functional consequence: loss of function via loss of secretion.** *"Computer simulations and deglycosylation assays indicate that the disease-causing founder variant p.Cys49Arg structurall[y]"* alters the protein such that mature, glycosylated WFDC2 is not secreted (PMID: 38626355). The net effect is **deficiency of a secreted airway component**.
- **Origin: germline**, biallelic (recessive).
- **Allele frequency:** pathogenic alleles are rare; the founder p.Cys49Arg suggests population-specific recurrence. (Exact gnomAD frequencies not extracted in this investigation.)

Modifier Genes

Not established for BENP. For the shared CRSwNP/AERD phenotype, arachidonic-acid-pathway and type-2 immune loci modulate severity (see Mechanism).

Epigenetic Information

Not established for BENP. Of note, in cystic fibrosis airway epithelium, *WFDC2*/HE4 mRNA is upregulated with decreased miR-140-5p, indicating HE4 expression is under microRNA regulation (PMID: 27105680) — relevant to WFDC2 biology though not to BENP pathogenesis per se.

Chromosomal Abnormalities

None reported; BENP is a single-gene disorder, not a structural/aneuploidy syndrome.

Environmental Information

- **Environmental factors:** Not primary. Chronic airway microbial exposure is the permissive environmental substrate on which the genetic defect acts.
- **Lifestyle factors:** No established BENP-specific lifestyle drivers. General bronchiectasis worsens with smoking and environmental exposures (extrapolated).
- **Infectious agents:** Chronic bacterial airway infection is central to the downstream pathology. In bronchiectasis broadly, microbiome dysbiosis with increased pathogenic bacteria correlates with severity (PMID: 42051198); older/advanced bronchiectasis shows a shift toward *Pseudomonas aeruginosa* and Enterobacteriaceae (PMID: 42474633). *Staphylococcus aureus* enterotoxins can amplify type-2 airway inflammation along the nasobronchial axis (PMID: 42646747).

Mechanism / Pathophysiology

Ordered Causal Chain

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1. Biallelic loss-of-function WFDC2 variant (e.g., p.Cys49Arg)          [demonstrated]
        │  leads to
2. Misglycosylation and blocked secretion of mature WFDC2 protein        [demonstrated]
        │  results in
3. Deficiency of secreted WFDC2 in airway surface liquid                 [demonstrated]
   (undetectable in serum)
        │  removes
4. A protease-inhibitor / antibacterial innate-defense component          [demonstrated/known]
   of the airway
        │  permits
5. Chronic bacterial airway infection and microbiome dysbiosis            [inferred/known]
        │  triggers
6. Persistent airway inflammation (neutrophilic ± type-2/eosinophilic)    [inferred/known]
        │  branches:
        ├──(UPPER AIRWAY)── epithelial injury + type-2 inflammation → mucus
        │    hypersecretion (dyscrinia) + tissue remodeling → recurrent NASAL
        │    POLYPOSIS, rhinosinusitis, nasal broadening                     [observed]
        │
        └──(LOWER AIRWAY)── impaired mucociliary clearance + neutrophil
             elastase / DPP-1-driven inflammation → "vicious vortex" →
             irreversible bronchial wall damage → BRONCHIECTASIS             [observed]
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Steps 1–4 are **demonstrated** by the primary genetic/biochemical study (PMID: 38626355). Steps 5–7 integrate established airway-disease mechanisms and are partly **inferred** for BENP specifically from the known biology of nasal polyposis and bronchiectasis.

Molecular Pathways and Cellular Processes

- **Upstream — loss of airway defense.** WFDC2/HE4 is a secreted serine-protease inhibitor. Curated GO annotations for WFDC2 (Gene 10406) include **serine-type endopeptidase inhibitor activity (GO:0004867)**, **peptidase inhibitor activity (GO:0030414)**, **negative regulation of endopeptidase activity (GO:0010951)**, **antibacterial humoral response (GO:0019731)**, **innate immune response (GO:0045087)**, **club cell differentiation (GO:0060486)**, and **positive regulation of lung ciliated cell differentiation (GO:1901248)**. Localization: **extracellular region (GO:0005576)** and **extracellular exosome (GO:0070062)** — consistent with a secreted protein. Loss of this protease-inhibitory/antibacterial shield underlies the "loss-of-airway-defense" mechanism.

- **Downstream (upper airway) — type-2 inflammation and remodeling.** Epithelial alarmins (IL-25, IL-33, TSLP) activate ILC2/Th2 cells → IL-4/IL-5/IL-13 → eosinophilia, local IgE, mast-cell activation: *"initial allergen exposure disrupts epithelial integrity, triggering local inflammation via alarmins including IL-25, IL-33, and TSLP, which activate type 2 innate lymphoid cells as well as other immune cells to secrete type 2 cytokines IL-4, IL-5 and IL-13, promoting Th2 cell development and eosinophil recruitment"* (PMID: 39158477). Eosinophil–epithelial interactions drive mucin and profibrotic cytokine secretion: *"Eosinophil-epithelial interactions significantly stimulated the secretion of MUC5AC, PDGF-AB, VEGF, TGF-beta1, and IL-8 in culture supernatants"* (PMID: 24717945) — the molecular basis of mucus hypersecretion (dyscrinia) and polyp remodeling.
- **Downstream (lower airway) — vicious vortex.** *"Bronchiectasis is a chronic respiratory disease characterised by irreversible bronchial dilatation, persistent airway inflammation and recurrent infections"* (PMID: 42342265). Neutrophil-driven inflammation via DPP-1/neutrophil elastase perpetuates airway damage; targeting DPP-1 (brensocatib) reduces exacerbations (PMID: 42048140).

Immune System Involvement

Combined **innate immunodeficiency** (loss of secreted WFDC2 antibacterial defense; GO:0045087) plus **chronic mixed inflammation** — type-2/eosinophilic in the polyp compartment and neutrophilic in the bronchiectatic compartment. A subset overlaps with AERD/Samter's triad, driven by dysregulated arachidonic-acid metabolism: *"Alterations in arachidonic acid metabolism may induce an imbalance between pro-inflammatory and anti-inflammatory substances, expressed as an overproduction of cysteinyl leukotrienes and an underproduction of prostaglandin E2"* (PMID: 29414455). Samter's triad was the most common comorbidity in reported Woakes' cases (7/39) (PMID: 41416018).

Cell Types and Tissues

- **Airway secretory/club cells (CL:0000158)** and **submucosal gland cells** — normal source of WFDC2.
- **Respiratory epithelial cells (CL:0002633)** and **ciliated cells (CL:0005012)** — mucociliary interface.
- **Eosinophils (CL:0000771)**, **mast cells (CL:0000097)**, **type-2 innate lymphoid cells / Th2 cells (CL:0000899)** — polyp inflammation.
- **Neutrophils (CL:0000775)** — bronchiectasis inflammation.

GO biological processes: GO:0019731 (antibacterial humoral response), GO:0045087 (innate immune response), GO:0002376 (immune system process), GO:0060486 (club cell differentiation), GO:1901248 (positive regulation of lung ciliated cell differentiation).

Molecular Profiling

WFDC2/HE4 is measurable in serum and airway. In BENP it is **undetectable in serum**, enabling diagnosis. By contrast, in cystic fibrosis, serum HE4 is *elevated* and correlates with disease severity — median 99.5 pmol/L in children with CF vs 36.3 pmol/L in controls ($P < .0001$) (PMID: 27105680) — illustrating that HE4 is dynamically regulated in airway disease and that its *absence* is the specific BENP signature.

Anatomical Structures Affected

Organ Level

- **Primary organs:** paranasal sinuses and nasal cavity (UBERON:0001707 nasal cavity; UBERON:0001825 paranasal sinus; frontal sinus UBERON:0002264) and the lungs/bronchi (UBERON:0002185 bronchus; UBERON:0002048 lung).
- **Secondary involvement:** middle ear (eosinophilic otitis media reported with Woakes', PMID: 27143164); nasal external framework (nasal broadening).
- **Body system: respiratory system** (UBERON:0001004), upper and lower airways ("unified airway").

Tissue and Cell Level

- **Airway epithelium** (respiratory epithelium), **submucosal glands**, and inflammatory infiltrate.
- Cell populations: club/secretory cells, ciliated cells, eosinophils, neutrophils, mast cells (CL terms above).

Subcellular Level

- **Extracellular region / airway surface liquid** (GO:0005576) — site of WFDC2 deficiency.
- **Secretory pathway / ER–Golgi glycosylation machinery** — where the p.Cys49Arg defect blocks maturation and secretion.

Localization / Lateralization

- Sinonasal disease is typically **bilateral**. Bronchiectasis in comparable pediatric cohorts preferentially involves **lower lobes** (left-lower 53.9–71%, right-lower 47–59%) (PMID: 30961955, PMID: 29605210).

Temporal Development

- **Onset:** variable — classically **early childhood** for the nasal-polyp component (Woakes'), but the systematic review documents onset from age 5 to adulthood (mean age at report 39.5 years) (PMID: 41416018). Onset pattern is **chronic/insidious**.
- **Progression:** chronic and **progressive** for bronchiectasis (irreversible bronchial dilatation) and **relapsing/recurrent** for nasal polyposis. Bronchiectasis follows a self-perpetuating "vicious vortex" (PMID: 42474633).
- **Duration: chronic, lifelong.**
- **Remission patterns:** treatment-induced partial remission (surgery, biologics) is common but polyps frequently recur; spontaneous remission is not characteristic.
- **Critical periods:** childhood diagnosis offers a window to institute airway-clearance and infection-control measures before irreversible bronchiectatic damage accrues.

Inheritance and Population

- **Inheritance pattern: autosomal recessive** (biallelic *WFDC2* loss-of-function).
- **Penetrance:** presumed high/complete for biallelic loss-of-function, though variable expressivity of individual phenotype components is expected.
- **Founder effects:** the recurrent **p.Cys49Arg** variant indicates a founder allele; affected families span the US, Europe, Asia, and Africa (PMID: 38626355).
- **Consanguinity:** relevant for a recessive disorder; high consanguinity rates characterize pediatric non-CF bronchiectasis cohorts generally (59.4%, PMID: 29605210).
- **Carrier frequency:** not precisely established; expected rare.

Epidemiology

BENP is **ultra-rare**: the defining series comprised 11 individuals from 10 families. The clinically overlapping Woakes' phenotype is similarly rare — a PRISMA systematic review identified only **39 unique patients across 23 studies** (*"Twenty-three studies met the inclusion criteria, comprising 39 unique patients (mean age 39.5 years; range 5-81; 65% male)"*, PMID: 41416018). Precise prevalence/incidence figures are not available.

Population Demographics

- **Sex ratio:** male-predominant in the Woakes' aggregate (65% male).
- **Geographic distribution:** multi-ancestry (worldwide) with a founder allele. In a Vietnamese sinopulmonary cohort, *WFDC2* abnormalities were specifically screened but **not identified**, underscoring rarity and possible population variation (PMID: 41094464).

Diagnostics

Clinical Tests

- **Serum HE4 (WFDC2) assay:** the key BENP-specific test — **WFDC2 is undetectable in serum** of affected individuals (contrast: elevated in CF). Combined with genetic testing, this enables diagnosis (PMID: 38626355, PMID: 27105680).
- **Imaging:** chest **HRCT** documents bronchiectasis (broncho-arterial ratio >1, lack of tapering, bronchial wall thickening, mucus plugging, tree-in-bud) (PMID: 31508157); sinus **CT** scored by Lund-Mackay; nasal endoscopy scored by Lund-Kennedy.
- **Biopsy/pathology:** nasal polyp histology shows eosinophilic mucosa with MUC5AC-rich mucus hypersecretion and remodeling.

Genetic Testing

- **Recommended approach:** targeted single-gene *WFDC2* sequencing, or a **sinopulmonary/ bronchiectasis gene panel** (should include *CFTR*, PCD genes *DNAI1/DNAH5/DNAH11*, and *WFDC2*), or **WES/WGS** for undiagnosed sinopulmonary disease. Vietnamese sinopulmonary genetic screening explicitly incorporated *WFDC2* alongside CF and PCD genes (PMID: 41094464).

Clinical Criteria and Differential Diagnosis

BENP/Woakes' diagnosis is partly **exclusionary**: childhood nasal polyposis is classically caused by CF or Kartagener/PCD — *"Usually, nasal polyposis in early childhood (children aged less than 5 years) is caused by cystic fibrosis of Kartagener's syndrome"* (PMID: 553887) — which must be excluded.

Differential	Gene(s)	Inheritance	Distinguishing tests
Cystic fibrosis	*CFTR*	AR	Sweat chloride ≥60 mmol/L; two pathogenic CFTR variants; **elevated** serum HE4 (PMID: 41919747, PMID: 27105680)
PCD / Kartagener	*DNAI1, DNAH5, DNAH11*	AR	Low nasal nitric oxide; ciliary EM; ± situs inversus (PMID: 16203616, PMID: 34391405)
AERD / Samter's triad	polygenic	—	NSAID hypersensitivity; aspirin challenge (PMID: 29414455)
BENP	*WFDC2*	AR	**Undetectable** serum HE4; biallelic WFDC2 variants; **normal** cilia and CFTR

For PCD exclusion: *"As a screening test nasal nitric oxide (NO) measurement is widely used. Establishment of diagnosis currently relies on electron microscopy, direct evaluation of ciliary beat by light microscopy"* (PMID: 16203616). The **absence** of serum HE4 (vs its elevation in CF) plus normal cilia distinguishes BENP.

Screening

Cascade genetic testing of at-risk relatives once a proband's biallelic *WFDC2* variants are identified; carrier testing for reproductive partners.

Outcome / Prognosis

- **Survival/mortality:** BENP is **not directly lethal**, but the bronchiectasis component drives long-term morbidity and mortality. In the European Bronchiectasis Registry (EMBARC, n=13,484), *"Sputum colour is a simple marker of disease severity and future risk of* *exacerbations, severe exacerbations and mortality in patients with bronchiectasis"* (PMID: 38609095).

- **Morbidity/QoL:** substantial — chronic nasal obstruction, anosmia, rhinorrhea (SNOT-22) plus chronic cough, sputum, dyspnea, and recurrent exacerbations.
- **Disease course:** chronic-relapsing. Sinus surgery relieves symptoms but *"endoscopic sinus surgery offers relief of symptoms and improvement of quality of life, recurrences are common and multiple surgeries are often required"* (PMID: 41261359).
- **Complications:** recurrent respiratory infections, progressive airflow obstruction, and (in overlap syndromes) eosinophilic otitis media.
- **Prognostic factors:** sputum purulence, *Pseudomonas* colonization, exacerbation frequency, and lung-function decline predict worse bronchiectasis outcomes (PMID: 38609095, PMID: 42474633).

Treatment

Surgical / Interventional

Functional endoscopic sinus surgery (FESS) is universal in reported cases — *"All patients underwent surgical treatment, most commonly functional endoscopic sinus surgery, with adjunctive procedures including digital nasal bone compression (six cases) and formal rhinoplasty or septorhinoplasty (seven cases)"* (PMID: 41416018). NCIT: Endoscopic Sinus Surgery.

Pharmacotherapy — Type-2-Targeted Biologics (for the polyp component)

Network meta-analyses of RCTs establish biologics as effective for the CRSwNP component:

Biologic	Target	Effect on Nasal Polyp Score (WMD vs placebo)	Source
Dupilumab	IL-4Rα (IL-4/ IL-13)	**−2.16 (95% CI −2.44 to −1.89)**	PMID: 41178615
Tezepelumab	TSLP	−1.50	PMID: 41178615
Mepolizumab	IL-5	~ −0.9 to −1.25	PMID: 41178615
Omalizumab	IgE	~ −0.9	PMID: 41178615

"NPS was significantly improved by dupilumab (WMD: -2.16, 95% CI [-2.44, -1.89])" (PMID: 41178615). Larger NMAs (3,642 patients, 7 biologics) rank **dupilumab, stapokibart, and tezepelumab** in the top efficacy tier (PMID: 42398863). A real-world cohort (n=360) achieved good-to-excellent response in **51%** (PMID: 41989130). NCIT: Dupilumab, Mepolizumab, Omalizumab.

Pharmacotherapy — Bronchiectasis Component

- **Airway clearance / mucoactive agents** (hypertonic saline, mannitol, carbocisteine); evidence is mixed — a meta-analysis did not show a significant reduction in exacerbations (PMID: 42342264).
- **Anti-inflammatory DPP-1 inhibition: brensocatib** (recently approved) reduces neutrophil-driven inflammation and exacerbations — *"the inhibition of dipeptidyl peptidase-1 (DPP-1) which can reduce neutrophil-driven airway inflammation and, consequently, exacerbations"* (PMID: 42048140). NCIT: Brensocatib.
- **Macrolides** (azithromycin) and targeted antibiotics for infection control.

Advanced / Experimental (Rational, Not Yet Available)

Because BENP is a **deficiency of a secreted protein**, it is conceptually amenable to **protein-replacement or gene-directed therapy** (inhaled/topical recombinant WFDC2, or *WFDC2* gene therapy) — analogous in spirit to CFTR modulation for CF. This is a proposed, not established, strategy.

Personalized Medicine

Genotype-guided care: confirmed *WFDC2* biallelic loss identifies patients for whom future WFDC2-directed therapy would apply and rationalizes aggressive combined upper/lower-airway management.

Prevention

- **Primary prevention:** not possible for a germline recessive disorder; **genetic counseling** and **carrier/cascade screening** in affected families are the key preventive tools.
- **Secondary prevention:** early diagnosis (serum HE4 + genetics) to institute airway-clearance and infection control **before** irreversible bronchiectasis develops (critical-period concept).
- **Tertiary prevention:** minimize exacerbations and structural progression via airway clearance, prompt antibiotics, DPP-1 inhibition, and polyp control (biologics/surgery).
- **Immunization:** routine respiratory immunizations (influenza, pneumococcal) are advisable to reduce infective exacerbations (standard bronchiectasis practice).
- **Counseling:** genetic counseling for autosomal-recessive recurrence risk (25% per pregnancy for carrier couples); prenatal/preimplantation options where variants are known.

Other Species / Natural Disease

- **Taxonomy / orthologs:** *WFDC2* is highly conserved with 1:1 mammalian orthologs — **mouse *Wfdc2* (Gene 67701), rat (286888), dog (403919), cow (618044), macaque (710469), chimpanzee (458283)**. NCBI Taxon: *Homo sapiens* (9606); *Mus musculus* (10090); *Rattus norvegicus* (10116).
- **Natural disease:** no naturally-occurring animal BENP homolog has been reported (OMIA search not positive in this investigation).
- **Evolutionary conservation:** strong ortholog conservation supports the feasibility of murine models and cross-species mechanistic study.

Model Organisms

- **Recommended primary model: mouse** *Wfdc2* knockout (Gene 67701) — the conserved 1:1 ortholog makes a targeted knockout the logical model to test whether WFDC2 loss recapitulates rhinosinusitis/nasal polyposis and bronchiectasis-like airway pathology with chronic infection.
- **Complementary in-vitro models: air–liquid interface (ALI) cultures** of human nasal/bronchial epithelium (used widely in CRS and CF research, e.g., PMID: 40683569, PMID: 27105680) with *WFDC2* knockdown/knockout to assay secretion, antibacterial activity, and mucin production; **patient-derived iPSC airway organoids**.
- **Model types available/needed:** knockout, conditional (airway-secretory-cell-specific), knock-in of p.Cys49Arg (to model the founder secretion defect), and humanized lines.
- **Phenotype recapitulation / limitations:** to be established — a key uncertainty is whether mice reproduce the human nasal-polyp phenotype (rodents rarely form true nasal polyps), which may limit face validity for the upper-airway component.
- **Resources:** MGI, IMPC/KOMP (for *Wfdc2* alleles), Cellosaurus/ATCC (epithelial lines).

Mechanistic Model / Interpretation

BENP is best understood as a **"loss-of-airway-defense" Mendelian disease**. A single secreted protein, WFDC2/HE4 — a WAP-domain protease inhibitor with antibacterial and protease-regulatory functions made by airway club cells and submucosal glands — is a non-redundant component of airway surface liquid. Biallelic loss (often via the p.Cys49Arg secretion-blocking founder variant) removes this defense, licensing chronic airway infection. The infection/inflammation then plays out along the **unified airway**: in the sinonasal compartment it drives type-2/eosinophilic inflammation, MUC5AC hypersecretion (dyscrinia), and remodeling → recurrent polyposis; in the bronchial compartment it drives the neutrophilic "vicious vortex" → irreversible bronchiectasis.

This model elegantly explains why BENP mimics CF and PCD (all three cause combined upper+lower airway suppurative disease) yet is molecularly distinct: **cilia are structurally normal (unlike PCD) and CFTR/chloride transport is normal (unlike CF)**. The serum HE4 test provides an unusually clean diagnostic discriminator — **absent in BENP, elevated in CF** — turning a rare genetic diagnosis into a two-step (blood biomarker → confirmatory sequencing) workflow.

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     UPSTREAM (genetic/molecular)          DOWNSTREAM (clinical)
WFDC2 biallelic LoF ──► no secreted    ──► chronic infection ──► ┬──► NASAL POLYPOSIS
(p.Cys49Arg etc.)      WFDC2 in ASL         + inflammation         └──► BRONCHIECTASIS
[demonstrated]         [demonstrated]       [inferred/known]            [observed]
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Evidence Base

PMID	Contribution	Weight
38626355	**Defining paper** — biallelic WFDC2 LoF causes BENP; founder p.Cys49Arg blocks secretion; serum HE4 undetectable	Landmark (human clinical + in vitro/computational)
27105680	HE4 biology in airway disease; elevated in CF (contrast with BENP); miR-140-5p regulation	Supporting
553887	Original Woakes' pentad; CF/PCD as differentials; hereditary basis	Historical/clinical
41416018	Systematic review — epidemiology (39 patients, 65% male), universal surgery, comorbidities	Aggregate clinical
25677865	Strong heritability of CRSwNP (4.1× first-degree relative risk; no spousal risk)	Supporting genetic basis
24717945	Eosinophil–epithelial MUC5AC/profibrotic mechanism (dyscrinia + remodeling)	Mechanism
39158477	Type-2 alarmin–cytokine cascade (upstream polyp inflammation)	Mechanism
42342265, 42048140, 42474633	Bronchiectasis "vicious vortex"; DPP-1 inhibition (brensocatib)	Mechanism/treatment
41178615, 42398863, 41989130	Biologic efficacy for CRSwNP (dupilumab leading)	Treatment
29414455	AERD/Samter's triad leukotriene mechanism (overlap endotype)	Mechanism
16203616, 34391405, 41919747	PCD and CF differential diagnosis	Diagnostics
38609095	Bronchiectasis prognosis (EMBARC registry)	Prognosis
29605210, 30961955, 40832796	Pediatric bronchiectasis etiology/lobar distribution	Context

PMID	Contribution	Weight
41094464	WFDC2 screening in a sinopulmonary cohort (not identified — rarity)	Epidemiology

Limitations and Knowledge Gaps

1. **Small defining cohort.** BENP is molecularly established from only 11 individuals in 10 families (PMID: 38626355); penetrance, expressivity, full variant spectrum, and natural history remain to be characterized in larger cohorts.
2. **Prevalence/carrier frequency undefined.** Precise gnomAD allele frequencies for pathogenic *WFDC2* variants and population carrier rates were not extracted here.
3. **Woakes'–BENP relationship not fully resolved.** How many historical Woakes' cases carry *WFDC2* variants is unknown; some Woakes'/CRSwNP-bronchiectasis cases are likely genetically heterogeneous (AERD, other loci). A Vietnamese cohort screened for *WFDC2* and found none (PMID: 41094464).
4. **Mechanistic steps 5–7 are partly inferred.** The infection→inflammation→structural-damage chain in BENP specifically is extrapolated from general airway-disease biology, not yet demonstrated in a BENP-specific model.
5. **No validated animal model** of BENP is yet reported; rodent nasal-polyp face-validity is a concern.
6. **Citation-integrity note.** Some citation snippets for the defining paper (P 38626355) were flagged as "mismatch" during automated validation; the quoted content is consistent with the paper's established conclusions but exact abstract wording should be re-verified before database ingestion.

Proposed Follow-up Experiments / Actions

1. **Confirm/curate ontology mappings** (OMIM:620984 ≡ MONDO:0975835 ≡ MedGen:1874999) and re-verify the exact abstract quotes from PMID: 38626355 to resolve the "mismatch"-flagged snippets before knowledge-base entry.
2. **Assemble a larger BENP case series** via GeneMatcher/international collaboration to define penetrance, expressivity, age-of-onset distribution, and the full *WFDC2* variant spectrum with gnomAD frequencies.
3. **Validate the two-step diagnostic algorithm** (serum HE4 → *WFDC2* sequencing) prospectively in unexplained combined rhinosinusitis-plus-bronchiectasis cohorts, quantifying sensitivity/specificity against CF and PCD.
4. **Generate a *Wfdc2*-knockout (and p.Cys49Arg knock-in) mouse** and *WFDC2*-null human ALI/organoid cultures; assay airway antibacterial activity, secretion, mucin production, and infection susceptibility to test causal steps 4–7.
5. **Assess WFDC2 protein-replacement or gene therapy** conceptually (recombinant WFDC2 in ALI/organoid rescue experiments) as a rational disease-modifying strategy.
6. **Retrospectively genotype archived Woakes'-syndrome cases** for *WFDC2* to quantify the molecular overlap between the historical clinical entity and the new Mendelian disease.

Report compiled from 12 confirmed findings across 5 investigation iterations and 60 reviewed papers. Evidence types are noted throughout as human clinical, in vitro, computational, or inferred/extrapolated.
